# Abstract


The **Casualty & Disaster Insurance Management System** is a full-stack enterprise-grade platform designed to automate and streamline the complete lifecycle of casualty and disaster insurance operations. The system manages policy creation, sales, claim submission, geographic disaster verification, fraud detection, investigation workflows, and claim settlement within a structured and secure environment

Built using **.NET 8 Clean Architecture, ASP.NET Core Web API, Entity Framework Core, SQL Server, and Angular with Tailwind CSS**, the platform ensures scalability, modularity, and maintainability through strict separation of Domain, Application, Infrastructure, and API layers

Unlike traditional insurance systems, this solution integrates advanced operational features such as:

- A **9-stage enforced claim workflow engine** that prevents invalid state transitions
- A **weighted fraud detection engine** with 10 rule-based risk assessments generating a fraud score (0–100)
- **Geographic disaster verification**, which cross-references incident location and date against disaster event data to validate claim authenticity
- **Role-based workflow separation** across five specialized roles (Admin, Sales Agent, Claims Manager, Claims Agent, and Customer)
- **Intelligent auto-approval logic** for low-risk, low-value claims
- Comprehensive **audit logging and commission tracking**

The system models real-world insurance company operations by incorporating fraud analytics, geo-verification, structured claim investigations, and automated payment processing. Through secure JWT-based authentication and role-based access control, it ensures operational transparency, accountability, and data integrity.

Overall, this project goes beyond a traditional CRUD-based insurance application by delivering a production-ready, workflow-driven insurance management ecosystem that mirrors the operational complexity of a mid-sized insurance enterprise